

19. Which of the following filter passes low frequencies?

- a) High pass filter b) Low pass filter
 c) Band pass filter d) None of the above

20. The spatial coordinates of a digital image (x,y) are proportional to:

- a) Position b) Brightness c) Contrast d) Noise

21. In general, the log transformation can be represented by _____

- a) $s = c \cdot \log(1 - r)$ b) $s = c - \log(1 - r)$
 c) $s = c \cdot \log(1 + r)$ d) $s = c + \log(1 + r)$

22. The principal disadvantage of piecewise functions is that their specification requires

- a) less user input b) adequate user input
 c) more user input d) significant user input

23. What is the full form of CAT in image processing?

- a) Computer-Aided Tomography
 b) Computer-Aided Telegaphy
 c) Computerized Axial Tomography
 d) Computerized Axial Telegaphy

24. What is meant by the section of the real plane that the image coordinates have spanned?

- a) Coordinate Axis c) Plane of Symmetry
 d) Spatial Domain d) None of the above

25. What is the name of the process in which the known data is utilized to evaluate the value at an unknown location?

- a) Acquisition b) Pixelation c) Interpolation
 d) None of the above

26. How to carry out an array function together with one or more images?

- a) Column by Column b) Array by Array
 c) Pixel by Pixel d) Row by Row

27. Which of the following the general representation of power transformation?

- a) $c = sr^r$ b) $s = rc^r$ c) $s = cr^r$ d) $s = rc$

28. What is the name of the tool that helps in zooming, shrinking, rotating, etc.?

- a) Filters b) Sampling c) Interpolation d) None of the above

29. The random variation of brightness or color information in images is _____

- a) Brightness b) Saturation c) Noise d) Intensity

30. Name the procedure in which individual pixel values of the digital image get altered.

- a) Neighborhood Operations b) Image Registration
 c) Single Pixel Operation d) Geometric Spatial Transformation

Structured Questions

1. State any five applications of digital image processing

5 marks

2. With the aid of a diagram, explain the image acquisition process 10 marks

3. With the aid of a diagram, explain the image sampling and quantization 10 marks

The End

NAME STIMFUKWE CHANCE

REG # MSE/20/85/032

Computer Graphics and Application - MTS-CGA-221 Fundamentals of Digital Image Processing QUIZ

1. Which of the following is the dominant application of microwave in image processing?
a) Radar b) medicines
c) lens enhancement d) medical diagnoses
2. Which of the following color is having largest frequency in visible spectrum?
a) Violet b) blue c) green d) red
3. Which of the following is the first fundamental step in image processing?
a) Filtration b) image restoration
c) image enhancement d) image acquisition
4. Which of the following rays have the maximum frequency?
a) UV Rays b) Radio Waves
c) Microwaves d) Gamma Rays
5. Which of the following is the third step in digital image processing?
a) Segmentation b) Image Restoration
c) Image Enhancement d) All of the above
6. What is called Digitizing the coordinate values of a continuous image?
a) Quantization b) Sampling
c) Compression d) Segmentation
7. Which of the following is the principal factor, which helps in determining an image's spatial resolution?
a) Contrast b) Sampling
c) Quantization d) Dynamic range
8. At what points, a continuous image is digitized?
a) Sampling b) Vertex
c) Contour d) Random
9. _____ represents the transition between image function's continuous values and its digital equivalent.
a) Quantization b) Rasterization
c) Sampling d) None of the above
10. Which of the following correctly describes the slightest visible change in the level of intensity?
a) Intensity Resolution b) Contour
c) Saturation d) Contrast
11. What is meant by Region of Interest (ROI) operations?
a) Dilation b) Shading correction
c) Masking d) None of the above
12. Which of the following is used to resolve the dark features in the image?
a) Gaussian Transform b) Laplacian Transform
c) Power-law Transformation d) Histogram Specification
13. Which of the following image component abruptly diverges at a certain junction of distinct objects?
a) Reflectance component b) Illumination component
c) Both (a) and (b) d) None of the above
14. Given an intensity level $[0, L-1]$ with "r" and "s" positive values, how will the negative of an image obtain?
a) $s = L - 1 + r$ b) $s = L + 1 - r$
c) $s = L - 1 - r$ d) $s = L + 1 + r$
15. What is the name of the process, which reverses the image's intensity?
a) Piecewise Linear Transformations
b) Log Transformations c) Image Negatives
d) None of the above
16. Which of the following is an image file format?
a) .ppt b) .avi c) .gif d) .mpg
17. What is the full form of JPEG?
a) Joint Photographs Expansion Group
b) Joint Photographic Expansion Group
c) Joint Photographic Experts Group
d) Joint Photographic Expanded Group
18. is a colour attribute that describes a pure colour.
a) Intensity b) Hue c) Brightness d) Saturation

Question one

25 Marks

- a. Define the following terms: **5 marks**
 - i. Computer graphics
 - ii. Shadow mask
 - iii. Noninterlaced system
 - iv. Rasterization
 - v. Framebuffer depth
- b. The four major areas of computer graphics are: display of information, user interfaces, design and simulation. With an example, describe how each area is used **8 marks**
- c. Describe two parts which distinguish computer graphics system from ordinary computer system **4 marks**
- d. Explain two properties of a raster images **4 marks**
- e. With the help of examples, explain the difference between lossless data compression and lossy data compression. **4 marks**

Question two

25 Marks

- a. Describe the color model used in OpenGL computer graphics **2 marks**
- b. Explain alpha blending in computer graphics **2 marks**
- c. With examples, explain the three basic kinds of geometric transform used in OpenGL **3 marks**
- d. With the aid of examples, explain the three OpenGL graphics libraries **3 marks**
- e. Describe any two characteristics of frame buffer that affect graphics performance **4 marks**
- f. Define a computer graphics rendering pipeline **1 mark**
- g. Outline the main stages in 3D graphics pipeline **4 marks**
- h. Describe OpenGL depth test **2 marks**
- i. Differentiate between illumination and shading **2 marks**
- j. State two general standards developed for graphics software **2 marks**

Question three

25 Marks

- a) How do you understand Pattern recognition? **2 marks**
- b) Describe the difference between Statistical pattern recognition and Structural pattern recognition **4 marks**
- c) Describe two key ideas on which Support Vector Machines learning algorithms are based **4 marks**
- d) Differentiate between Convolutional neural networks (CNNs) and Recurrent neural networks (RNNs) **4 marks**
- e) State any three classifiers used in computer vision **3 marks**
- f) Describe two strengths of neural networks learning **4 marks**
- g) Explain any two machine vision applications **4 marks**

Question four

25 Marks

- a. Define a digital image **1 mark**
- b. State the first two fundamental steps in image processing **2 marks**
- c. Describe three application areas of image processing with appropriate examples. **6 marks**
- d. Explain the Image acquisition process **6 marks**
- e. With the aid of a diagram, illustrate how a model of the image degradation / restoration process occurs **4 Marks**
- f. Differentiate the illumination and reflectance of a digital image formation **2 marks**
- g. Distinguish the following terms
 - i. Sampling and quantization **2 marks**
 - ii. spatial domain and frequency domain methods. **2 marks**

The End

Bachelor of Mathematics and Statistics
[MTS]
Year 2
MID SEMESTER TEST

Computer Graphics and Application COM 208

INSTRUCTION(S): Answer ALL questions

1. Define the following: **(1 mark each)**
 - a. Computer graphics
 - b. refresh rate
 - c. rasterization
 - d. Pixel
 - e. frame-buffer
2. With examples distinguish the following **(2 marks each)**
 - a. window port & view port
 - b. Raster and vector images
 - c. RGBA and HSLA
 - d. Local and infinite light sources
 - e. ambient reflection and specular reflection
3. Discuss the three shading models used in computer graphics **(9 marks)**
4. With the aid of examples, explain the three OpenGL graphics libraries **(6 marks)**
5. Explain three OpenGL basic transformations **(6 marks)** *rotate, translate, scaling*
6. Use Figure 1 attached to answer the following questions:
 - i. Why is it important to include `#include<windows.h>` and `#include<GL/glut.h>` libraries? **(2 marks)**
 - ii. Explain each statement in the main function **(1 mark each)**
 - iii. Describe the role of each of these gl functions in the display function:
`glClearColor ( )`, `glClear ( )`, `glBegin ( )`, `glColor3f ( )`, `glVertex2f ( )`, `glEnd ( )`, `glFlush ( )`

Handwritten notes:
Handwritten
Solid


```

1  #include<windows.h>
2  #include<GL/glut.h>
3  void display() {
4      glClearColor(0.0f, 0.0f, 0.0f, 1.0f);
5      glClear(GL_COLOR_BUFFER_BIT);
6
7      glBegin(GL_TRIANGLE_STRIP);
8          glColor3f(1.0f, 0.0f, 0.0f);
9          glVertex2f(-0.5f, -0.5f);
10         glVertex2f(0.5f, -0.5f);
11         glVertex2f(0.5f, 0.5f);
12         glVertex2f(-0.5f, 0.5f);
13     glEnd();
14     glFlush();
15 }
16 int main(int argc, char** argv) {
17     glutInit(&argc, argv);
18     glutCreateWindow("OpenGL Setup Test");
19     glutInitWindowSize(320, 320);
20     glutInitWindowPosition(50, 50);
21     glutDisplayFunc(display);
22     glutMainLoop();
23     return 0;
24 }

```

Figure 1: OpenGL program